

Track 2: Java Basics

Exercise 1: Writing Java Codes 1

Instruction: This is a hands-on seatwork. Follow the instructions given below.

Hours: 3.5 hours

Name your project as **<FamilyName>Ex1**

- A. Correct the following codes by **fixing some bugs** and applying the **Java coding standards**. Drop all your corrected classes inside **org.ssglobal.training.codes.a**.

Erroneous Code 1

```
public class Oops {
    public static void main(String[] args) {
        int a = 7, b = 42;
        int smaller = minimum(a, b);
        if {smaller = a} {
            System.out.println("a is the smallest!");
        }else{
        }
    }

    public static void minimum(int a, int b) {
        // returns which int is smaller

        if (a < b) {
            int smaller = a;
        } else (a => b) {
            int smaller = b;
        }
        return int smaller;
    }
}
```

Erroneous Code 2

```
public class LotsOf Errors {
    public static main(String args) {
        System.println("Hello, world!");
        message()
    }

    public static void message {
        System.out println("This program surely cannot";
        System.out.println("have any so-called "errors" in it");
    }
}
```

Erroneous Code 3

```
public class Temperature {
```



```

        exercise e = new exercise();
        int x = 200;
        e.f(x);
    }
}

```

Erroneous Code 7

```

public class Tricky
{
    public static main(String args) {
        System.out.println(Hello world);
        system.out.Pritnln("Do you like this program?");
        System.out.println()

        System.println("I wrote it myself.");
    }
}

```

Erroneous Code 8

```

public class Addition {
    public static void main(String[] args) {
        int x;
        System.out.println("x is" x);

        int x = 15.2;    // set x to 15.2
        System.out.println("x is now + x");

        int y;           // set y to 1 more than x
        y = int x + 1;
        System.out.println("x and y are " + x + and + y);
    }
}

```

- B. Write a Java class **AssignMe** that has an entry point **main()** which performs the following steps:
- Declare two int variables named **x** and **y**.
 - Assign 3 to **x**.
 - Assign twice the value of **x** to **y**.
 - Interchange the value of **x** and **y** (without explicitly assign 3 to **y**).
 - Print the values of both variables on screen.

Drop the class inside **org.ssglobal.training.codes.b**.

- C. Write a Java class **DeclareMe** that has an entry point **main()** which performs the following steps:
- Declare a float variable called **f**.
 - Declare an int variable called **k**.
 - Assign 22 to **k**.
 - Assign the value of **k** to **f**.
 - Print the values of both variables on screen.

Drop the class inside **org.ssglobal.training.codes.c**.

D. Study the following problem scenario:

Before implementing your solution in Java syntax, write the pseudocode of a class **SalesTax** that has a method **computeCost(itemCost)** that will compute the amount of state sales tax due, and the amount of county sales tax due. The state sales tax rate is constant at **4%**. The county sales tax rate is constant at **2%**.

The program should then display:

- The cost of the item purchased
- The amount of state sales tax due
- The amount of county sales tax due
- The total amount due from the user (cost of item + amount of state sales tax + amount of county sales tax)

Create a test class **TestSalesTax** to run **computeCost(itemCost)** with varying **itemCost**.

Do the following:

1. Applying the flowcharting discusses in **Track 1**, create your initial solution in a flowchart form using **draw.io**. Save the PNG format of your flowchart in **src/pseudocodes**.
2. Implement your flowchart using Java syntax discussed today from declaration of variables, choosing the right data types, and writing the proper codes. Place your solution inside **org.ssglobal.training.codes.d**.

E. The following program is legal under Java's syntax rules, but it is **difficult to read** because of its layout and **lack of comments**. Reformat it using the rules given in this chapter and add a comment header at the top of the program regarding what the class is supposed to do. Place the class inside **org.ssglobal.training.codes.e**.

```

/*****
// Place here your description
*****/
public class SumProd { public static void main(String[] args){
final int INT2=8; final int INT1=20; System.out.println(
"The sum of " + INT1 + " and "
+ INT2 + " is " + (INT1+INT2)); System.out.println (
"Their product is " + (INT1*INT2)); }}
```